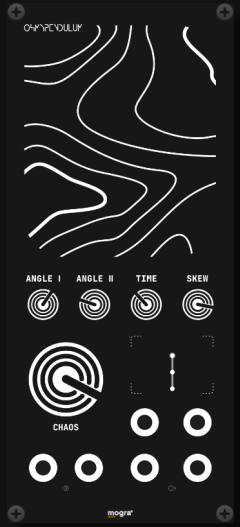
OHMYMODULAR

Carefully crafted VCV Rack modules with a twist.

made with love & chaos by

mogra audio × keithrosseau

OLYPENDULUM



MONO



CREME



MINT



MAGE

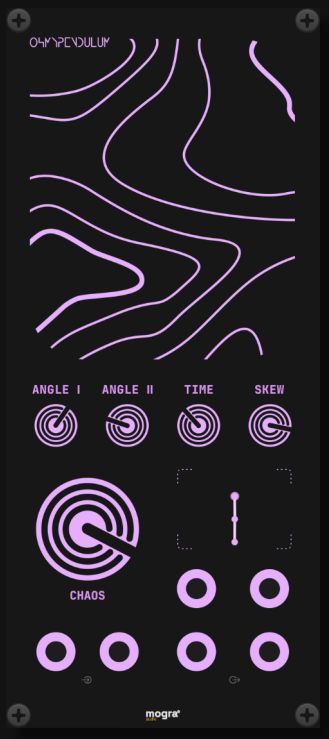


LOVE

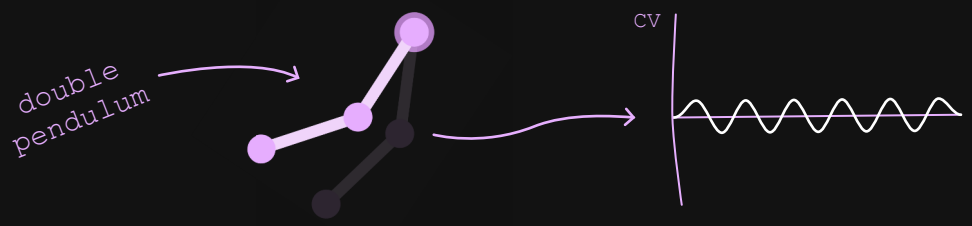
Chaotic double pendulum CV generator with a twist.

OHMYPENDULUM

OhMyPendulum(OMP) is a chaotic cv generator module for **VCV Rack**. Based on a double pendulum(DP) simulation, it transforms its angular motion to control voltage(CV) values. Each arm provides a separate cv value.



Unlike traditional LFOs or random sources, **OMP** produces mathematically-rich generation derived from real-time physics simulation using the 4th-order Runge-Kutta numeric integration method (RK4). The panel also includes a live animated DP drawn directly on the module. The module design was carefully handcrafted with precision for your comfort.



FEATURES:

- Real-time physics simulation & animation of a DP system
- Conservative energy for constant motion & generation
- 5 control knobs for initial conditions and behavior shaping
- 2 cv inputs for external modulations
- 4 cv outputs — smooth & discrete for each pendulum's arm
- Chaos knob — drives the simulation crazy
- Expander module to extend the possibilities

USER MANUAL

OhMyPendulum lets you explore a vast spectrum of creative patching possibilities: from subtle, drifting modulations to unpredictable chaotic bursts. There is no single "correct" way to use it — its voice is yours to shape.

The simulation is based on Hamiltonian mechanics and integrated with a 4th-order Runge-Kutta method for high accuracy. Angles are scaled to ±5 V so they can directly modulate any CV input in your patch. The module occupies 11 HP. All controls are clearly laid out, and the panel features an animated DP that follows the simulation in real time.
You can also switch between five visual themes from the right-click context menu: Mono, Creme, Minty, Mage, Love.

To get started, simply set the angle knobs to your desired value. A moment after you release the knob, the simulation springs to life, carrying the pendulum into motion.

MODULE PARAMETERS:

ANGLE I: sets the initial angle (0-360°) of the top DP's arm.

TIME: adjusts the time scale of the simulation. Center (12 o'clock) = real time. Turn left to slow down (dwn to 0.2×), turn right to speed up (up to 7×).

CHAOS: recursive parameter that uses a "ghost" DP to modulate the main one in the background. It pushes the simulation into an extreme condition & induces even more chaotic behavior. This allows you to get more sophisticated output voltage patterns.

ANGLE II: sets the initial angle (0-360°) of the bottom DP's arm.

SKEW: shifts the mass distribution between the two arms. Center = equal. Far left = heavy top (1.5 kg/0.5 kg), far right = heavy bottom (0.5 kg/1.5 kg).

MODULE INs:

RETRIG IN: resets the simulation using an external rising edge signal (thrshld > 1 V). Parameter values are preserved on reset.

CHAOS IN: CV control over the **Chaos** amount. The voltage controls the knob value (0–5 V maps to the full knob range).

OUTs:

(all outputs are bipolar & safe within ±5 V)

ANGLE I (Smooth): continuous angle of the top arm (±5 V).

ANGLE I (Discrete): quantized to 1/12 V steps (semitone grid) of the top arm (±5 V).

ANGLE II (Smooth): continuous angle of the bottom arm (±5 V).

ANGLE II (Discrete): quantized to 1/12 V steps (semitone grid) of the bottom arm (±5 V).

OHMYPENDULUM EXPANDER

OhMyPendulum Expander allows you to extend the possibilities of **OMP** module by providing you with extra inputs and outputs. The module occupies 6 HP. It activates when you drag it to the right of OMP instance and it automatically inherits its theme. It adds CV control over TIME & SKEW and exposes the Cartesian (XY) positions of the pendulum arms as four outputs. It has no audio path of its own; it reads the state published by the **OMP**.
When CV is present at the expander inputs, the knobs on **OMP** become inactive and tint gray.



CARTESIAN OUTs:

(technical information)

The expander converts the two DP angles into Cartesian coordinates and exposes the position of the DP as four control voltages. The pivot is the origin of the XY plane, with X pointing right and Y pointing up. Both arms have the same normalized length, $L_1 = L_2 = 1.0$. The angles θ_1 and θ_2 are measured from the downward vertical: $\theta = 0$ points straight down, and positive angles rotate toward +X. Crucially, θ_2 is the absolute angle of the second arm, not an angle relative to the first arm. The elbow — the joint between the two arms — is at $x_1 = L_1 \cdot \sin(\theta_1)$, $y_1 = -L_1 \cdot \cos(\theta_1)$. The tip of the second arm is at $x_2 = x_1 + L_2 \cdot \sin(\theta_2)$, $y_2 = y_1 - L_2 \cdot \cos(\theta_2)$. The negative cosine places the arms below the pivot at rest. The four outputs are Angle I – X = $2.5 \cdot x_1$, Angle I – Y = $2.5 \cdot y_1$, Angle II – X = $2.5 \cdot x_2$, and Angle II – Y = $2.5 \cdot y_2$, where Angle I describes the elbow and Angle II describes the tip. The same scale of 2.5 V per unit length is used for both points, so an XY display shows the true geometry rather than two independently normalized shapes. The elbow therefore stays within ±2.5 V, while the tip can reach ±5 V; all outputs are clamped to ±5 V as a safety limit, and non-finite values are output as 0 V. The XY outputs are not slew-limited, because they should follow the actual pendulum position.

MODULE INs:

TIME IN: CV control over the **TIME** amount. The voltage controls the full knob range value (0–10 V).

SKEW IN: CV control over the **SKEW** amount. The voltage controls the full knob range value (0–10 V).

OUTs:

(all outputs are bipolar & safe within ±5 V)

ANGLE I (X): continuous cartesian elbow X (top arm) (±2.5 V).

ANGLE I (Y): continuous cartesian elbow Y (top arm) (±2.5 V).

ANGLE II (X): continuous cartesian tip X (bottom arm) (±5 V).

ANGLE II (Y): continuous cartesian tip Y (bottom arm) (±5 V).

QUICK PATCH GUIDE:

1. Set both **angle knobs** to your desired starting position.

2. Patch **ANGLE I** or **ANGLE II** outputs to the **V/Oct** input of any VCO. You can also add your own quantizer (e.g VCV QNT) in between to choose a special musical mode & scale.

3. Patch any trigger module (e.g VCV PUSH) to **Retrigger** input so you can restart the simulation whenever you want.

4. Adjust **TIME** & **SKEW** to change the character. Add a little **CHAOS** for extra wildness.

USE CASES:

- **Modulation source** — smooth outputs can modulate filter cutoff, waveshape or panning. Slow the simulation down with **Time** knob for gentle, evolving sweeps.

- **Rhythmic triggers** — create fascinating polyrhythms with a few instances of **OMP** modules: Quantize the **OMP**'s full output range (**2πrad**) with a module of your choice. Use any CV to Trigger module (e.g Repelzen Re-Trig) to generate signal and feed it into any VCO or other modules.

- **Generative melodies** — use the discrete outputs (or quantize the outputs yourself) to drive any VCO.

- **Chaos bursts** — crank the **CHAOS** knob, then reset with a trigger input to create short bursts of extreme modulation.

- **Modulate** **TIME** CV and **SKEW** CV (with **OMP Expander**) from external sources or recursively to change the pendulum's speed and mass distribution.

- **Patch** ANGLE II - X/Y as a correlated stereo pair for panning, filter cutoff, or any two parameters that should move together geometrically.

Enjoy exploring the chaotic beauty of OMPENDULUM!

OHMYMODULAR



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